The witness ending one step after the crossing

Human Compression project: L044 continuation

All marks are involutions satisfying string commutations. The local actions below need no intersection property.

Lemma 1 (One right mark)**.** *Let $Q = \langle \rho_m, \dots, \rho_{j+1} \rangle$ be transitive on W, with every displayed mark active. Suppose the full local j-omission has two orbits joined by exactly one j-edge, the first omission $Q_{>m}$ is intransitive, and*

$$h = j - m \geq 0, \qquad |W| = h + 3.$$

Then $Q_{>m}$ has precisely two orbits, a singleton and its transitive complement.

Proof. Put $L = Q_{<j}$ and $R = \langle \rho_{j+1} \rangle$. The actual omission sides A, B are transitive under LR, with crossing endpoints a, b.

First suppose $h \geq 3$. The sharper acquired commuting-factor bound excludes attaining the left length h and right length one on the same side: that side alone would require $h + 3$ points. Thus, after naming the sides,

$$|A| = h + 1, \quad |B| = 2, \quad L|_A = \mathrm{Sym}(A), \quad R|_A = 1, \quad R|_B = \mathrm{Sym}(B). \tag{1}$$

The same conclusion holds for $h = 2$, unless both maxima lie on a four-point side A, with the other side a singleton. We exclude that exception explicitly. The single involution R cannot be transitive on A, and the earlier commuting-factor count would give six points if neither factor were transitive. Hence L is transitive on A. Write $t_0 = \rho_{j-1}|_A$, $t_1 = \rho_m|_A$, and $c = \rho_{j+1}|_A$. The involution c is a nontrivial centralizer of the transitive L, hence is fixed-point-free. The mark t_1 fixes a, and therefore also a^c. Being nonidentity, it is the transposition of the other two points. Transitivity of L forces t_0 to exchange the two c-pairs. Commutation with c then makes t_0 a product of two cross-pair transpositions. Consequently $\langle t_0, c \rangle$ is transitive on the four points. These marks survive the first omission, as does the crossing to the singleton. The first omission would be transitive, a contradiction. This proves (1) for $h = 2$ also.

Under (1), every left mark except ρ_{j-1} fixes b, because it commutes with ρ_j and preserves the unique crossing. It thus fixes both points of B. The nearest left mark may also transpose B, but R itself supplies that transposition. On A, the exhausted anchored path is $a_0 = a, \dots, a_h$, in reverse left order. The only possible extra path transposition is the acquired decoration $(a_{h-1}\ a_h)$ in the mark three positions from the end. If present, the remaining left marks after omitting ρ_m would still connect all of A. Together with R and the crossing they would make the first omission transitive. Hence that decoration is absent, and its remaining left marks connect exactly $A \setminus \{a_h\}$ and fix a_h.

For $h \geq 3$, the far left marks give the pure $\mathrm{Sym}(\{a_1, \dots, a_h\})$, fixing B. Commuting with this group makes ρ_j fix its entire support. For $h = 2$, the only possible additional action of ρ_j on A is $(a_1\ a_2)$; it would connect the first omission and is therefore absent. Thus $z = a_h$ is fixed by the first omission, whose other points are connected by the remaining left action, the crossing and R. This gives the assertion for all $h \geq 2$.

If $h = 1$, there are four points. An omission side has transitive action of the commuting involutions L, R, so its size is a power of two. Thus the sides each have size two. The unique crossing is the literal transposition of their chosen endpoints. If R transposed both sides, $\langle \rho_j, R \rangle$ would be transitive on all four points, contrary to the first-omission hypothesis. It must transpose

exactly one side. Its crossing therefore gives a transitive three-point orbit and fixes the unused point of the other side. If $h = 0$, there are three points, $L = 1$, and the active involution R is a transposition. It is itself the first omission, proving the last case. □

Lemma 2 (An anchor in the left endpoint orbits)**.** *Let $Q = \langle \rho_m, \dots, \rho_y \rangle$ be transitive on W, every mark active, with a unique full local j-crossing $\{a, b\}$ and $y \geq j+2$. Put $h = j - m \geq 0$, $L = Q_{<j}$, $R = Q_{>j}$ and*

$$C = a^L \sqcup b^L.$$

Suppose $|C| = h + 2$ and some $z \in C$ is fixed by the first omission $Q_{>m}$. Then that omission has precisely the orbits $\{z\}$ and $W \setminus \{z\}$.

Proof. For $h \geq 1$, the anchored bound and equality make one of these two L-orbits a set A_0 of size $h + 1$, with $L|_{A_0} = \mathrm{Sym}(A_0)$, and the other a singleton b, after renaming. Since ρ_j belongs to the first omission and exchanges a, b, the fixed point z lies in $A_0 \setminus \{a\}$. The whole right factor fixes z and centralizes L, so fixes every point of $A_0 = z^L$. Thus the complete actual omission side through a is $A = A_0$. The left factor fixes b and centralizes R, so the other actual side is $B = b^R$ and L fixes all of B.

For $h \geq 2$, the exhausted anchored path $a_0 = a, \dots, a_h$ shows that the surviving left marks connect a_0 through a_{h-1}. Thus their fixed point $z \neq a_0$ can only be a_h; a possible extra edge joining a_h is excluded by the assumed fixation of z. For $h = 1$, the set $A \setminus \{z\}$ is just $\{a\}$. In every case the surviving left marks are transitive on $A \setminus \{z\}$. The right factor is transitive on B, and ρ_j joins these sets while fixing z. This proves the conclusion.

If $h = 0$, then $C = \{a, b\}$ and the first omission is R. Its fixed anchor z is one of a, b. The actual omission side through z is therefore a singleton; R is transitive on the other side. This proves this case too. □

Retain the complete actual odd-degree head and paired-tail hypotheses, and the strong simultaneous choice of crossing colour $j \geq 5$. Write $H = G_{\geq 5}$, $K = G_{>5}$, $\sigma = \rho_4$. Every K-intransitive witness orbit has first active label 5, a unique full local crossing, and pointwise fixation by δ in the actual head interface $\rho_3 = (6\ b)\delta$.

Theorem 3. *Let O_1, O_2 be two distinct K-intransitive witness orbits, with last active labels $x = j + 1 < y$. Then $x < r - 1$ and*

$$|\operatorname{Fix}(G_{>x})| \geq 2x + 1.$$

Proof. The acquired simultaneous-witness count gives $x < r - 1$ and at least $2x$ fixed points. If the asserted stronger bound fails, the fixed set $F = \operatorname{Fix}(G_{>x})$ is precisely

$$F = D \cup \{b\} \ \sqcup\ O_1 \ \sqcup\ C, \qquad |O_1| = x - 3, \quad |C| = x - 4, \tag{2}$$

where C is the union of the two left endpoint orbits in O_2. In particular $|C| = h + 2$ for $h = j - 5$, and the first lemma applies on O_1. Its K-orbits are a singleton $\{z_1\}$ and a transitive complement T_1, with $|T_1| = |C|$.

The actual anchor $z_0 = b^\sigma$ is outside the seven head points and is fixed by K. It lies in F, so (2) places it either at z_1 or in C. The involution σ centralizes K, hence permutes its actual orbits. It also preserves F, since $x \geq 6$.

Whenever the K-orbits of a witness O_i are $\{z_0\}, T_i$, the involution σ cannot preserve T_i. Otherwise the proper set $D \cup \{b\} \cup O_i$ would be invariant under every generator: the first three marks preserve D; ρ_3 exchanges $6, b$ and fixes O_i through δ; σ exchanges b, z_0, fixes D and preserves T_i; and H preserves O_i and fixes the seven head points. The other witness is outside this set, contrary to transitivity of G.

Suppose first $z_0 = z_1$. The distinct K-orbit T_1^σ lies in F. It cannot meet the fixed head points or z_1, and cannot equal T_1. Equation (2) therefore puts it inside C. Their equal cardinalities give $T_1^\sigma = C$, so C is K-invariant. It is also invariant under ρ_5: for $j > 5$ this mark belongs to L, whose

two orbits define C; for $j = 5$, C consists of the two crossing endpoints, interchanged by ρ_5. Thus C is H-invariant. It must equal the whole actual orbit O_2. But $C \subset F$ and ρ_y is active on O_2, with $y > x$, a contradiction.

Suppose instead $z_0 \in C$. The second lemma makes the K-orbits on O_2 exactly $\{z_0\}, T_2$. By the preceding invariant-set argument σ moves T_2 to a distinct K-orbit. Take any point of $C \setminus \{z_0\}$; this set is nonempty, since $|C| = x - 4 \geq 2$. Its image is in F and in the nontrivial orbit T_2^σ. It cannot be one of the K-fixed points, and cannot lie in O_2, whose only nontrivial K-orbit is T_2. By (2) it lies in T_1, so $T_2^\sigma = T_1$. This is impossible: $\rho_y \in K$ is nontrivial on T_2 but fixes T_1 pointwise, while σ centralizes it. Both cases contradict the assumption $|F| = 2x$, proving the bound. □

Attribution and boundary. The actual head interface, simultaneous witness selection, anchored paths and orbit counts are acquired. The two local equality arguments and the action of the head anchor on the entire fixed set are supplied here. All short cases are included. The case $x = j$, a second witness with transitive K-action and full outside attachments remain. No whole-original, question, formal or numerical credit is claimed.